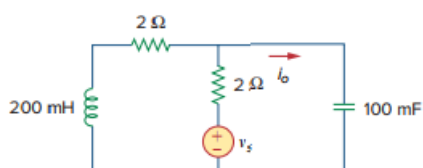


Problem

If $v_s = 100 \sin(10t + 18^\circ)$ V in the circuit of Fig. 9.53, find i_o .

Figure 9.53:



Step-by-step solution

Step 1 of 3

Refer to Figure 9.53 in the text book for the circuit.

The source voltage $v_s(t)$ is,

$$\begin{aligned} v_s(t) &= 100 \sin(10t + 18^\circ) \text{ V} \\ &= 100 \cos(10t - 72^\circ) \text{ V} \end{aligned}$$

The angular velocity ω is,

$$\omega = 10 \text{ rad/s}$$

Write the source voltage in phasor domain.

$$\mathbf{V} = 100 \angle -72^\circ \text{ V}$$

Determine the capacitive impedance, $\mathbf{Z}_C$.

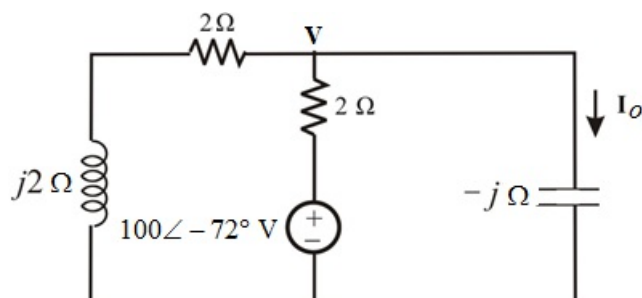
$$\begin{aligned} \mathbf{Z}_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(10)(100 \times 10^{-3})} \\ &= -j \Omega \end{aligned}$$

Determine the inductive impedance, $\mathbf{Z}_L$.

$$\begin{aligned} \mathbf{Z}_L &= j\omega L \\ &= j(10)(200 \times 10^{-3}) \\ &= j2 \Omega \end{aligned}$$

Step 2 of 3

Draw the circuit in frequency domain.



Step 3 of 3

Apply Kirchhoff's current law at node $\mathbf{V}$.

$$\frac{\mathbf{V}}{2+j2} + \frac{\mathbf{V}-100\angle-72^\circ}{2} + \frac{\mathbf{V}}{-j} = 0$$

$$\mathbf{V}\left(\frac{1}{2+j2} + \frac{1}{2} + \frac{1}{-j}\right) = \frac{100\angle-72^\circ}{2}$$

$$\mathbf{V}(0.75 + j0.75) = 50\angle-72^\circ$$

$$\mathbf{V} = 47.14\angle-117^\circ \text{ V}$$

Determine the current, $\mathbf{I}_o$.

$$\begin{aligned}\mathbf{I}_o &= \frac{\mathbf{V}}{\mathbf{Z}_c} \\ &= \frac{47.14\angle-117^\circ}{-j} \\ &= 47.14\angle-27^\circ \text{ A}\end{aligned}$$

Convert the current $\mathbf{I}_o$ into time domain.

$$\begin{aligned}i_o(t) &= 47.14\cos(10t-27^\circ) \text{ A} \\ &= 47.14\sin(10t+63^\circ) \text{ A}\end{aligned}$$

Thus, the current $i_o(t)$ is $47.14\sin(10t+63^\circ) \text{ A}$.

[Comments \(1\)](#)



Anonymous

How do you get from $(1/(2+j2) + 1/2 + 1/-j)$ to $(0.75 + j0.75)$?
